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Appendix S1: Supplementary Text

A. Data set details. The agroforestry Experiment: three years of data from a cashew(*Anacardium occidentale*) monoculture intercropped with intensively managed groundnuts.

B. Data set modifications. Data set modifications: Data set modifications: Data quality assurance measures, controlling plant variability, spatial effects, time effects

C. Metadata description.

A. Data set details

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C. Metadata description.

B. Data set modifications

Data quality assurance measures, controlling plant variability, spatial effects, time effects

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11G1	Initial sown proportion of fast-establishing grass	
12G2	Initial sown proportion of persistent grass	
13L1	Initial sown proportion of fast-establishing legume	
14L2	Initial sown proportion of persistent legume	
15E	Initial sown evenness	
16DENS	Indicator variable: high=high level of initial sown biomass, low = low level (60%of high)	
17G1_Y	Harvest Dry Matter Yield of fast-establishing grass	t/ha
18G2_Y	Harvest Dry Matter Yield of persistent grass	t/ha
19L1_Y	Harvest Dry Matter Yield of fast-establishing legume	t/ha
20L2_Y	Harvest Dry Matter Yield of persistent legume	t/ha
21WEED_Y	Harvest Dry Matter Yield of weed species	t/ha
22HARV_YIELD	Total Harvest Dry Matter Yield	t/ha

Column numbers, variable names, and variable descriptions for file: forage_quality.csv

Column	Variable Name	Variable Description	Unit
1	SITE	Site ID number	
2	country	Country	
3	YEAR	Year	
4	YEARN	Experimental year number	
5	NH	Number of harvests	
6	Harvest	Harvest number (within year)	
7	HARVEST_DATE	Date of harvest	
8	PLOT	Plot number as per design (1–30 = core design; 31–48 = treatment plots; 49–68 = additional plots at sites 45 and 46)	
9	TREAT	Indicator variable: 1=basic 30 plots; 2 and 3=additional treatment plots (some sites implemented two levels of additional treatments)	
10	REP	Replicate number (applies only to sites 15 and 45)	
11	G1	Initial sown proportion of fast-establishing grass	
12	G2	Initial sown proportion of persistent grass	
13	L1	Initial sown proportion of fast-establishing legume	
14	L2	Initial sown proportion of persistent legume	
15	E	Initial sown evenness	
16	DENS	Indicator variable: high=high level of initial sown biomass, low = low level (60% of high)	
17	Local_N	Indicator variable (Local lab analysis present=1, absent=0)	
18	Kiel	Indicator variable (Kiel bulk sample present=1 , absent=0)	
19	Canada	Indicator variable (Canada bulk sample present=1 , absent=0)	
20	Kiel_sep	Indicator variable (Kiel separated sample present=1 , absent=0)	
21	N_L	Nitrogen percent in total harvest yield (analysis performed by local lab)	% of dry matter
22	N_K	Nitrogen percent in total harvest yield (Kiel data)	% of dry matter
23	Ash_K	Ash in total harvest yield (Kiel data)	% of dry matter
24	NDF_K	Neutral Detergent Fibre in total harvest yield (Kiel data)	% of dry matter
25	ADF_K	Acid Detergent Fibre in total harvest yield (Kiel data)	% of dry matter
26	CDOMD_K	Cellulase Digestible of Organic Matter of Dry Matter in total harvest yield (Kiel data)	% of dry matter
27	ME_K	Metabolizable Energy in total harvest yield (Kiel data)	MJ ME per kg of DM
28	N_C	Nitrogen percent in total harvest yield (Canadian data)	% of dry matter
29	NDF_C	Neutral Detergent Fibre in total harvest yield (Canadian data)	% of dry matter
30	ADF_C	Acid Detergent Fibre in total harvest yield (Canadian data)	% of dry matter
31	IVTD_C	In Vitro True Digestibility in total harvest yield (Canadian data)	
32	IVCWD_C	In Vitro Cell Wall Digestibility in total harvest yield (Canadian data)	
33	N_G1_K	Nitrogen percent in G1 harvest yield (Kiel data)	% of dry matter
34	Ash_G1_K	Ash in total G1 harvest yield (Kiel data)	% of dry matter
35	NDF_G1_K	Neutral Detergent Fibre in total G1 harvest yield (Kiel data)	% of dry matter
36	ADF_G1_K	Acid Detergent Fibre in total G1 harvest yield (Kiel data)	% of dry matter
37	CDOMD_G1_K	Cellulase Digestible of Organic Matter of Dry Matter in G1 harvest yield (Kiel data)	% of dry matter
38	ME_G1_K	Metabolizable Energy in G1 harvest yield (Kiel data)	MJ ME per kg of DM
39	N_G2_K	Nitrogen percent in G2 harvest yield (Kiel data)	% of dry matter
40	Ash_G2_K	Ash in G2 harvest yield (Kiel data)	% of dry matter
41	NDF_G2_K	Neutral Detergent Fibre in G2 harvest yield (Kiel data)	% of dry matter
42	ADF_G2_K	Acid Detergent Fibre in G2 harvest yield (Kiel data)	% of dry matter
43	CDOMD_G2_K	Cellulase Digestible of Organic Matter of Dry Matter in G2 harvest yield (Kiel data)	% of dry matter
44	ME_G2_K	Metabolizable Energy in G2 harvest yield (Kiel data)	MJ ME per kg of DM
45	N_L1_K	Nitrogen percent in L1 harvest yield (Kiel data)	% of dry matter
46	Ash_L1_K	Ash in L1 harvest yield (Kiel data)	% of dry matter
47	NDF_L1_K	Neutral Detergent Fibre in L1 harvest yield (Kiel data)	% of dry matter
48	ADF_L1_K	Acid Detergent Fibre in L1 harvest yield (Kiel data)	% of dry matter
49	CDOMD_L1_K	Cellulase Digestible of Organic Matter of Dry Matter in L1 harvest yield (Kiel data)	% of dry matter
50	ME_L1_K	Metabolizable Energy in L1 harvest yield (Kiel data)	MJ ME per kg of DM
51	N_L2_K	Nitrogen percent in L2 harvest yield (Kiel data)	% of dry matter
52	Ash_L2_K	Ash in L2 harvest yield (Kiel data)	% of dry matter
53	NDF_L2_K	Neutral Detergent Fibre in L2 harvest yield (Kiel data)	% of dry matter
54	ADF_L2_K	Acid Detergent Fibre in L2 harvest yield (Kiel data)	% of dry matter
55	CDOMD_L2_K	Cellulase Digestible of Organic Matter of Dry Matter in L2 harvest yield (Kiel data)	% of dry matter
56	ME_L2_K	Metabolizable Energy in L2 harvest yield (Kiel data)	MJ ME per kg of DM
57	N_weed_K	Nitrogen percent in weed species harvest yield (Kiel data)	% of dry matter
58	Ash_weed_K	Ash in weed species harvest yield (Kiel data)	% of dry matter
59	NDF_weed_K	Neutral Detergent Fibre in weed species harvest yield (Kiel data)	% of dry matter
60	ADF_weed_K	Acid Detergent Fibre in weed species harvest yield (Kiel data)	% of dry matter
61	CDOMD_weed_K	Cellulase Digestible of Organic Matter of Dry Matter in weed species harvest yield (Kiel data)	% of dry matter
62	ME_weed_K	Metabolizable Energy in weed species harvest yield (Kiel data)	MJ ME per kg of DM

DATA-USE POLICY

The data presented here are publicly available. Those wishing to publish results from this data set should read this metadata document. The data set should be cited as: Kirwan et al. 2014. The Agrodiversity Experiment: three years of data from a multisite plant diversity experiment in intensively managed grasslands. Ecology xx:xxx.

Three papers are currently in preparation, focusing on (a) the effect of diversity on the contribution of weed species to biomass yield, (b) changes in the relative abundances of the mixtures, and (c) grassland biodiversity effects across an environmental gradient.

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